U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure 01-2-100 27 June 2025

3-DIMENSIONAL IMAGE ANALYSIS USING HIGH-SPEED CAMERAS

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1. SCOPE.

Digital high-speed imagery is used to capture transient events or fast-moving objects during testing. The captured imagery can be used to gain qualitative understanding of the test article or object in motion with respect to time. With prior planning and proper calibration of cameras, quantitative data can be extracted to determine motion dynamics of the test article including but not limited to position, velocity, acceleration, and angle. The test articles can range from small fragments or projectiles to larger targets such as full-sized vehicles, or vehicle components. By extracting quantitative data, stake holders can gain a more detailed understanding of the test article's motion and performance.

The objective of this Test Operations Procedure (TOP) is to cover three-dimensional (3D) image analysis techniques using fixed high-speed cameras to support test and evaluation (T&E) programs. This document is not intended to be a step-by-step guide to execute any one specific test, rather, it addresses ideal setups and analysis methods using high-speed camera imagery and diagnostics to measure Time-Space-Position Information (TSPI) of a test article. Since testing requirements vary greatly from test-to-test, any subjective descriptors in this document (e.g., too small, too slow) are all related to the quantity of interest for a specific test instead of a universal value. This TOP is beneficial in supporting groups such as test officers, engineers, and technicians, to understand our methods of conducting testing. TrackEye[1*] and ProAnalyst[2], two commercial off-the-shelf (COTS) software packages** predominantly used by the U.S. Army Test and Evaluation Command (ATEC) for 3D analysis, are also discussed in this document to clarify workflow path to deliverable data meeting test requirements.

2. FACILIITIES AND INSTRUMENTATION.

Facilities and instrumentation required to acquire high-speed imagery data will be dependent on the test requirements. High-speed data for 3D image analysis can be set up in an indoor or an outdoor facility as needed.

2.1 Facilities.

The test range conducting testing will follow local Standard Operating Procedures (SOPs) and TOPs. Should a discrepancy arise between this document and local procedures, local procedures should be followed.

*Superscript numbers correspond to Appendix B, References.
**The use of brand names does not constitute endorsement by the Army or any other agency of the Federal Government, nor does it imply that it is best suited for its intended application.

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2.2 Instrumentation.

The following instrumentation is typically required for high-speed camera setups for image acquisition and 3D analysis:

a. High-Speed Cameras. Camera pairs being used for acquisition should have similar specifications.

b. Trigger. Trigger signal generator to distribute time zero to all high-speed cameras for event synchronization.

c. Master Clock Source. Timing source to distribute common timestamps to all cameras. Inter-Range Instrumentation Group Format B (IRIG-B) and Global Positioning System (GPS) are two examples of time generators that provide micro-second or nano-second level accuracy.

d. Cabling and Network Equipment. Copper, fiber optics, and associated networking equipment to connect multiple cameras to instrumentation control center. Cabling is also required for synchronization, trigger, and timing signal.

e. Light Source. High intensity light-emitting diode (LED) lights, laser illuminators, flash bulbs, etc. Depending on test requirements and exposure duration, high-speed cameras may require artificial lighting to minimize distortion of the image from motion blur.

f. Calibration Fixtures. Flat grid board or other rigid fiducial fixtures used for calibrating camera parameters and defining the coordinate system.

g. Surveying. Geodetic surveying to support camera calibration by surveying location of cameras and calibration fixtures in the specified coordinate system for meeting test requirements.

3. REQUIRED TEST CONDITIONS.

Test conditions to consider will consist of environmental and technical elements.

3.1 Environments.

Outdoor testing will need to consider the weather conditions in which testing will occur. Rain and snow will affect the optical medium and will lead to distortion of the imagery. Tests should be delayed if heavy rain or snow is affecting camera views. Likewise extreme heat can have “mirage” effect in the image recorded. Extreme heat will also require proper instrumentation protection and cooling.

Indoor testing typically will require added lighting to properly expose the test article and to minimize motion blur ensuring more accurate location.

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3.2 Test Plan Requirement.

The test requirements need to be fully understood before testing to ensure the proper selection of high-speed camera instrumentation. Below are some general parameters to consider:

a. How small or large is the object of interest being tracked.

b. How large of an area the object of interest needs to be tracked for, field-of-view (FOV), in each dimension.

$$FOV\ horizontal = \frac{Horizontal\ sensor\ size * standoff\ distance}{Focal\ length}$$

$$FOV\ vertical = \frac{Vertical\ sensor\ size * standoff\ distance}{Focal\ length}$$

c. How fast is the object of interest moving and in what direction (frame rate).

$$Frame\ rate = \frac{Velocity\ of\ object}{Distance\ object\ travelled}$$

d. Maximum exposure time to eliminate motion blur.

$$Motion\ blur = Velocity\ of\ object\ \times exposure\ time$$

e. Test location (indoor/outdoor).

f. High-speed camera placement.

g. Instrumentation, safety, and protection.

3.3 Instrument Protection and Stabilization.

a. High-speed cameras require rigid mounting to prevent camera shake and proper protection from test events that may have fragmentation, fireballs, shockwaves, and vibrations. Camera protection can be achieved by placing cameras inside protective enclosures specifically designed for the test environment. Enclosures can be made from materials such steel, polycarbonate, and ballistic glass. Additionally, cameras can be shielded by concrete blocks, sandbags, or other structures and use a mirror to view the area of interest as opposed to having direct line-of-sight (LOS).

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b. Camera stabilization can be achieved by mounting cameras to heavy, rigid fixtures that do not move easily. Sandbags should be used on the tripod legs of the cameras to improve stabilization. In situations where 3D tracking is required well after an explosion or shock event, there may be perceived movement in the FOV due to movement of the camera instead of movement of the test article. Stabilization points in the FOV can aid stabilizing the recorded video in post-process image analysis. Stabilization points are features that are considered stationary and are not moved by the explosive event.

4. TEST PROCEDURES.

4.1 Background.

Foundational optical setup and calibration background are discussed in the following sections.

4.1.1 Testing Background.

a. As the main purpose behind the usage of the 3D image analysis techniques and associated equipment discussed is to acquire test data, the cameras are the main instrumentation for data acquisition. This is conceptually similar in function to a strain gauge in a traditional tensile test. However, as the raw data acquired by the cameras comes in the form of images, the data acquisition process for cameras differs greatly when compared to traditional methods. Knowledge of the camera's data acquisition process is necessary to understand the capabilities and limitations of a given test setup.

b. Overall, each camera records the test environment/event and transmits this data to a user interface, usually a computer. 3D real world data is transmitted through the camera's lens to its sensor plane, where it is recorded as a digital two-dimensional (2D) image. These 2D images are what the user interacts with as the basis for 3D image analysis.

c. In a general 3D image analysis setup, a minimum of two cameras are needed as a second camera view allows data pertaining to the depth of the test environment to be properly acquired. For wide field-of-view (WFOV) setups, it is recommended to have at least three cameras for redundancy. In setups where more than two cameras are being used, cameras are typically organized as multiple pairs of cameras. The cameras should be placed at a recommended offset ranging from 30 to 60° between each other, depending on the test situation. Angles above 90° or too less than 15° between cameras can present challenges in correlating common data points. In addition, small angle separation between cameras introduces larger errors in the depth estimation of the object being tracked. The general area where the desired event is occurring should be visible in both camera views. If further vertical FOV is desired, the high-speed cameras can be mounted on their side to get optimal sensor coverage since most high-speed cameras have a rectangular sensor with more horizontal pixels (i.e., 1920x800). Depending on the test requirements and/or range set up, there may be objects that cannot be removed from the FOV. This may lead to visual obstructions that have the potential to disrupt analysis. It is imperative that there is at least one high-speed camera pair that can see the object being tracked at all times of interest.

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d. In Figure 1, item a), the camera angles are roughly 60° and the FOV covers the point of interest. In Figure 1, item b), multiple camera pairs are in place to account for the visual obstruction. Depending on the specific test setup, it may be necessary for the object of interest to pass through another object. If data are needed on the other side of that object, additional camera pairs will need to be set up. A sample of a proper camera set up, with ideal angle of separation between cameras is shown in Figure 2, while an example of narrow angle of separation that may result in larger depth estimation errors is shown in Figure 3.

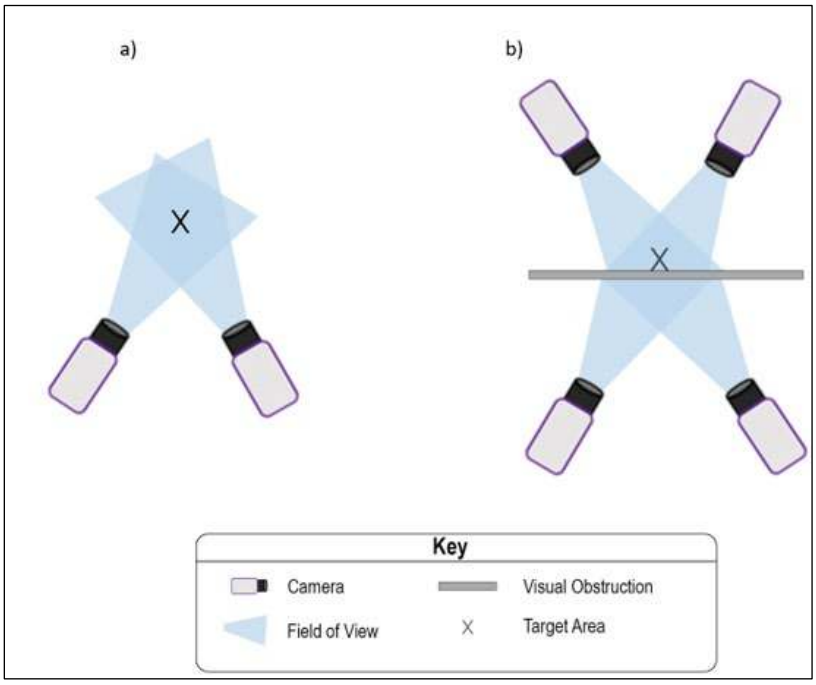


Figure 1. Ideal camera setup: a) setup for no visual obstruction of object, b) visual obstruction with a pair of cameras to see behind the target.

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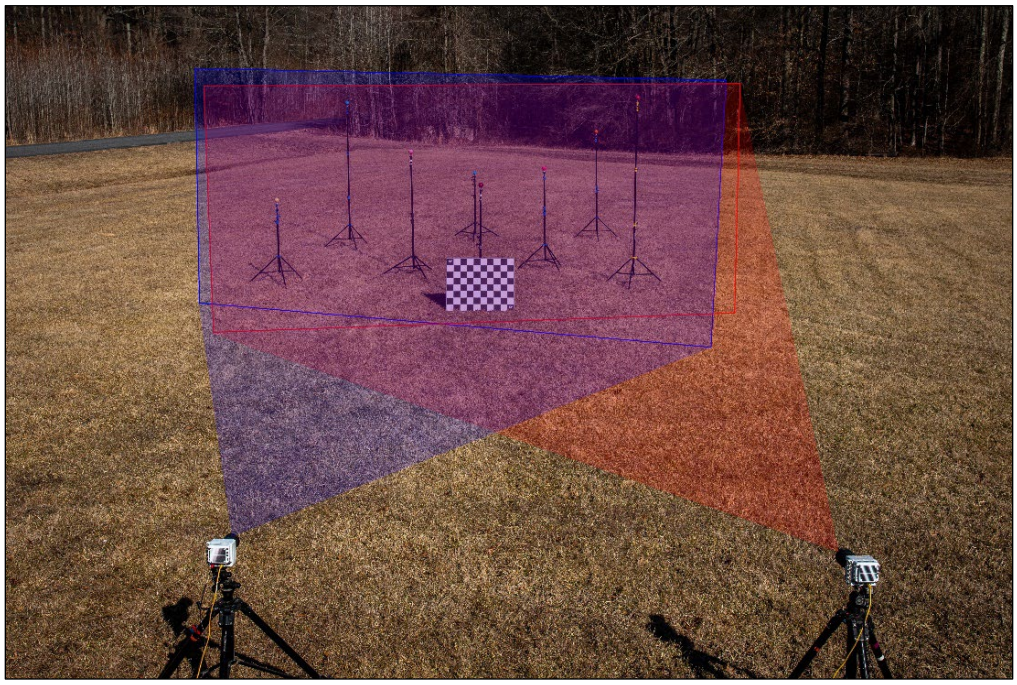


Figure 2. Representation of ideal camera set up with 60° camera separation.



Figure 3. Example of narrow camera set up with 15° camera separation.

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e. Each camera obtains different data sets as they output separate camera views of the same test environment. Since the data must correlate when analyzed, the two data sets must be synced with each other. This is typically performed temporally by defining a common point in time through a triggering device or master time clock. Further temporal accuracy can be obtained by syncing each frame from each camera. It is imperative to ensure properly synced cameras for data acquisition and analysis, not only for optical analysis but for comparison with other instrumentation such as radar and ballistics gauges that may be present on the test. Additionally, artificial lighting may be required based on test conditions.

4.1.2 Calibration Background.

a. Camera calibration is the crucial step that supplies the necessary information to transform the data from image coordinates to real-world coordinates. This is done by defining the camera's intrinsic and extrinsic parameters. Intrinsic parameters are properties of the camera contained within the camera itself, e.g., focal length, principal point, and skew. The extrinsic parameters are properties of the camera as it relates to the real world, e.g., position and orientation. The two primary methods for determining these parameters are calibrating for absolute orientation and calibrating for relative orientation.

b. Absolute orientation calibration (AOC) uses a world (range) coordinate system that is fixed. Relative orientation calibration (ROC) calibrates the scene in an arbitrary coordinate system, typically that of one of the cameras or the calibration fixture. For ROC, the coordinate system changes with different locations of the calibration fixture(s). Position and orientation are defined in both cases, thus defining the extrinsic parameters. The intrinsic parameters are independent of the coordinate system and thus defined similarly between the two methods.

c. AOC requires a series of calibration nodes with known x, y, z coordinates in the desired coordinate system, that can be seen in both camera views. ROC requires points that can be seen in both camera views, but since one of the cameras serves as the system origin, only one measurement needs to be known as opposed to the coordinates of all calibration points. Calibration procedures will be defined in a later section.

d. Neither AOC nor ROC corrects for lens distortions, which may contribute to inaccurate data in severe cases. An example of how distortion affects an image is shown in Figure 5. Based upon test requirements, a lens correction may be required for a lens with high distortion.

4.1.3 Error Sources.

There are many factors that may introduce error to optical measurements that should be noted, both outside of human control and due to human error. Since image analysis relies on optical data, anything that may impact the optical medium can introduce error.

4.1.3.1 Calibration.

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All sources of error and uncertainty begin with calibration. Improper calibrations will lead to errors in the data and even good calibrations have uncertainty measurements of their own. For example, if you are geodetically surveying your calibration nodes, your measurement can only be as accurate as the geodetic equipment being used to survey the calibration points. As 3D analysis requires a calibrated volume instead of a calibrated plane, error introduced from objects moving in and out of a 2D plane is accounted for. The caveat to this is if the test article begins to move out of the calibrated volume, as this will begin to increase the uncertainty the further the test article moves from the calibrated volume. In conjunction with proper spatial calibration, lens distortion correction should be considered. Lens distortion is caused by imperfections in the lens or by the physics of light rays passing through a curved surface. It usually presents as a barreling, pincushion, and tangential distortions of the imagery. Examples of these distortions are shown in Figure 5. When high precision near the edges of the image or small focal lengths is required, users should consider correcting for lens distortion.

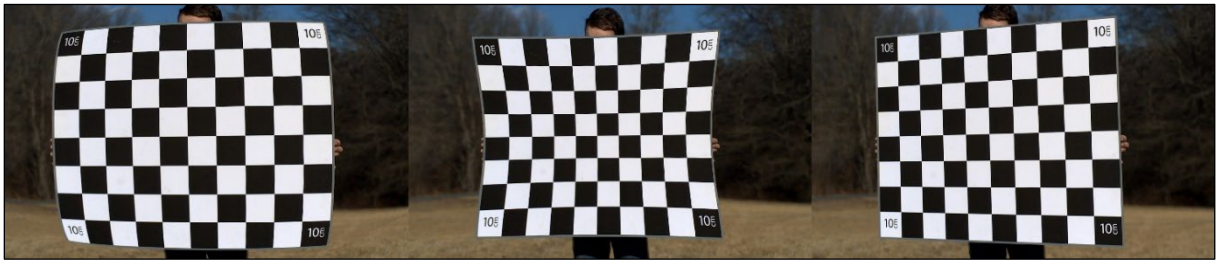


Figure 5. Examples of barrel distortion (left), pincushion distortion (middle), and tangential distortion (right).

4.1.3.2 Environmental Considerations.

Weather is one source that is not within the control of the test officers or camera operators. Precipitation can lead to disturbances or obstructions in the image. Excessive clouds can lead to poor lighting conditions which could lead to lack of contrast between the test article and its background. Since the temperature of the air affects how light travels through it, excessive heat can lead to heat distortions in the imagery which may lead to inaccurate tracking. These heat distortions often present as wavy lines or a shimmering effect within the imagery, an example of which is shown in Figure 6. Some of these can be mitigated with artificial lighting or moving a test indoors, however, these may not be feasible with a given test setup.

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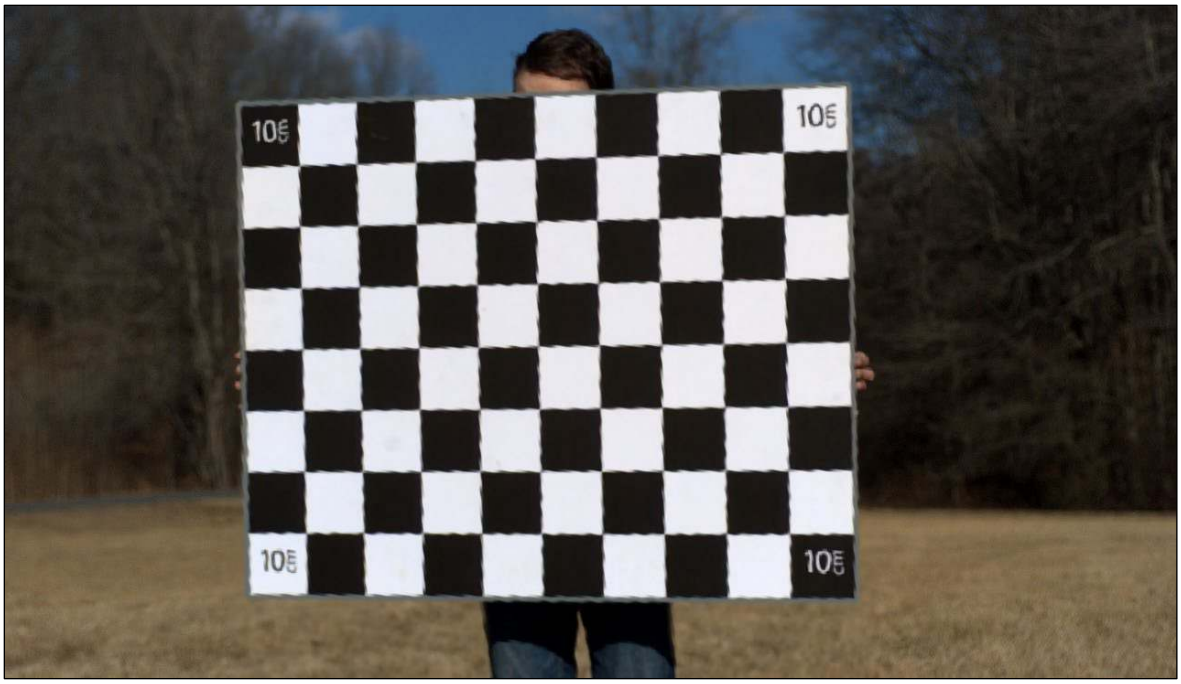


Figure 6. Example of heat distortion in an image.

4.1.3.3 Data Acquisition.

There are a variety of settings in the acquisition process that affect the amount of error in the result. If the frame rate is too low, respective to the velocity of the subject of interest, not enough data points may be sampled for a conclusive result. If the exposure is too long, the resulting motion blur may make features about interest difficult to distinguish, impacting the accuracy of the data. See equations in section 3.2 for details on frame rate and exposure. Depending on test requirements, frame syncing may be necessary to minimize the uncertainty in the time difference between camera frames.

4.2 Range Setup.

This section will cover range set-up considerations for various types of testing scenarios.

4.2.1 High-Speed Camera Setup for WFOV.

a. WFOV can be defined as a FOV roughly 10 by 10 m or greater. Some of the common test scenarios include vehicle motion analysis, airburst or impact testing, and 3D fragment tracking. Due to the large area covered, the calibration process will typically require surveying calibration fixtures as planar grid boards may not adequately cover such a FOV.

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b. Depending on the FOV required, three or more cameras may be necessary to ensure complete visual coverage. It is also good practice to have redundant calibrated views for cases where the test article may end up outside the expected FOV, or where one or more cameras are obscured or otherwise rendered unusable during the event.

4.2.2 High-Speed Camera Setup for Narrow Field-of-View (NFOV) or Indoor Testing.

NFOV can approximately be defined as a FOV less than 10 by 10 m. Some of the common test scenarios include small projectile tracking, impact scoring, behind armor debris tracking, helmet and other protective equipment impact testing. The calibration process will typically require a small calibration fixture or small planar calibration grid board. Since NFOV testing typically occurs in tighter areas, custom fixtures may be required to protect testing equipment. Artificial lighting may be required due to lack of natural lighting.

4.2.3 Detailed Example of a High-Speed Camera Setup for Active Protection System (APS) Analysis Containing both WFOV and NFOVs.

a. High-speed imaging is used in testing APS's ability to detect, track, engage, and neutralize an incoming threat. The optical instrumentation needed will be comprised of high-speed cameras, trajectory trackers, standard speed video cameras, still cameras, and high-intensity LED lighting or laser illuminators. The use of all or part of the instrumentation listed will vary depending on test requirement, system under test, and range configuration.

b. Typical APS testing will require a high-speed view at the threat launch, all or part of threat flight path, and the APS. High-speed camera views will also be required at Projected Intercept Point (PIP). All high-speed cameras should have a minimum 1-megapixel resolution to resolve smaller objects in the FOV.

c. To properly assess the performance of the APS, multiple camera views are required at the PIP. The high-speed views of the threat launch, and countermeasure launch are recorded between 5,000 to 7,000 frames per second (fps) for detailed view of first movement or light flash from the threat and countermeasure. The high-speed view of threat flight is recorded between 5,000 to 10,000 fps, orthogonal to the shot line. A trajectory tracker with a high-speed camera may also be used to track the threat from launch to intercept. One or more high-speed views on a tower, running at the same speed, may be included for an overhead view of the system.

d. A typical setup used at ATC for APS testing is shown in Figure 7. The PIP area will comprise several overlapping high-speed camera views for multi-angle 3D analysis. High-speed cameras will be positioned orthogonally to shot line of threat and countermeasure. Additional views at 45 and 15° to shot line, and tower views of the same overlapping area are to be positioned for 3D analysis. It is also good practice to have a redundant tight view on PIP area and a wider view in cases when intercept occurs before or after the expected PIP area.

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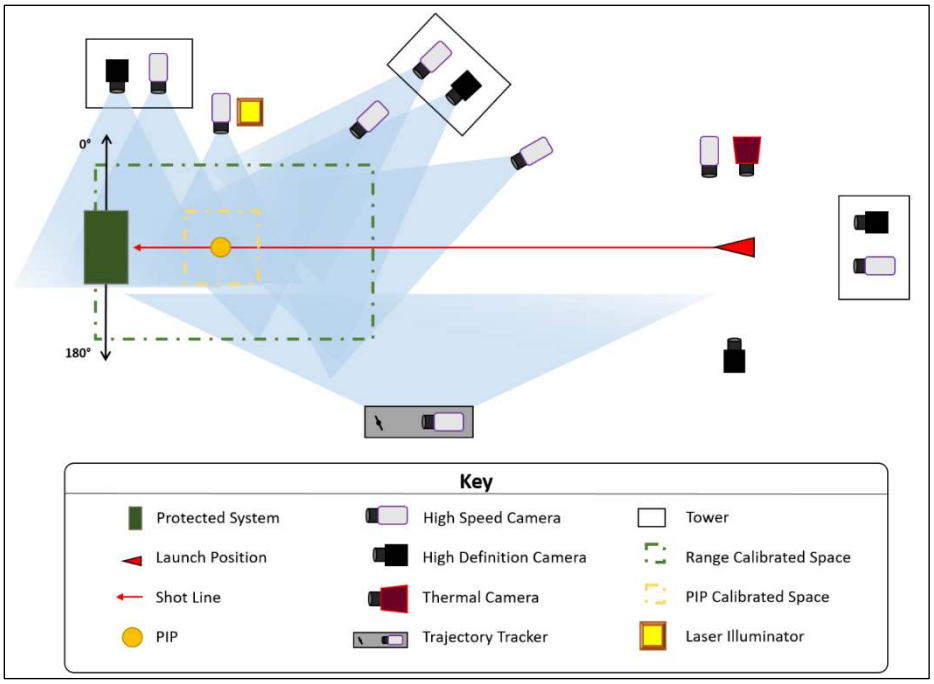


Figure 7. Example camera setup for APS test.

e. All cameras at the PIP should be framed synchronized for analysis and timing information. Frame rate typically will vary from 20,000 to 50,000 fps depending on the system under test. Velocities of threat and countermeasure should be used to calculate minimal motion blur of a few millimeters or less. Typical camera exposure time is 1µs or less. If ambient light is not sufficient to light the scene, additional illumination may be used such as high intensity LED lighting or laser illuminator with narrow banded wavelength that falls within camera sensor's sensitivity spectrum. In cases where explosive or incendiary events overexpose the sensor, a narrow band pass filter matching the wavelength of the laser illuminator may be attached on camera lenses to remove majority of the wavelength of light present while still illuminating the scene. All optical equipment should be contained in protective enclosures to prevent equipment damage during test event.

4.3 Camera Calibration Acquisition.

a. Calibration can broadly be broken down by two methods: planar and non-planar. Planar calibrations use a flat calibration board (often with a grid pattern) as shown in Figure 8. The pattern of a planar calibration board that uses circles instead of a grid pattern is shown in Figure 9. Non-planar calibration uses fiducial markers or fixtures in the FOV with known coordinates as shown in Figure 10.

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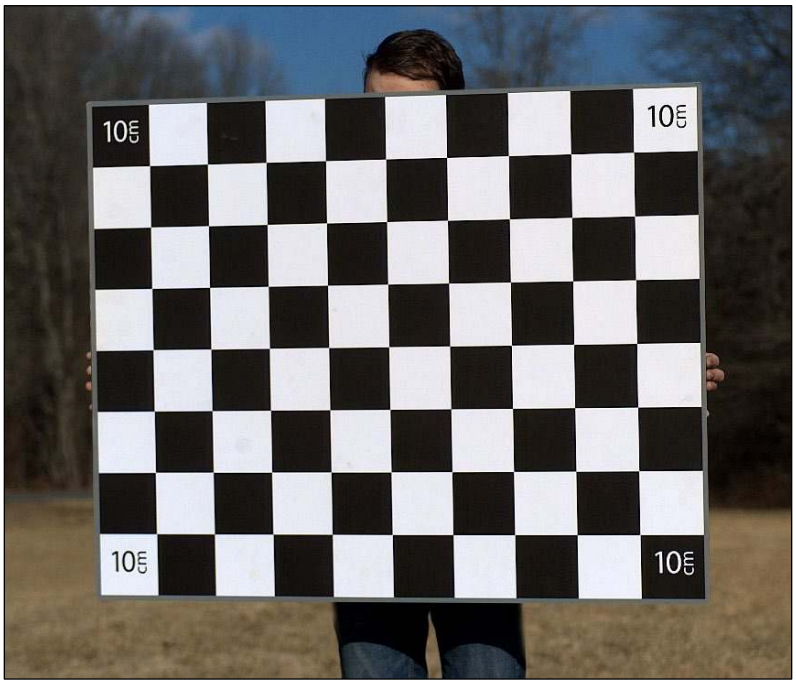


Figure 8. Example of checker-board style planar calibration board.

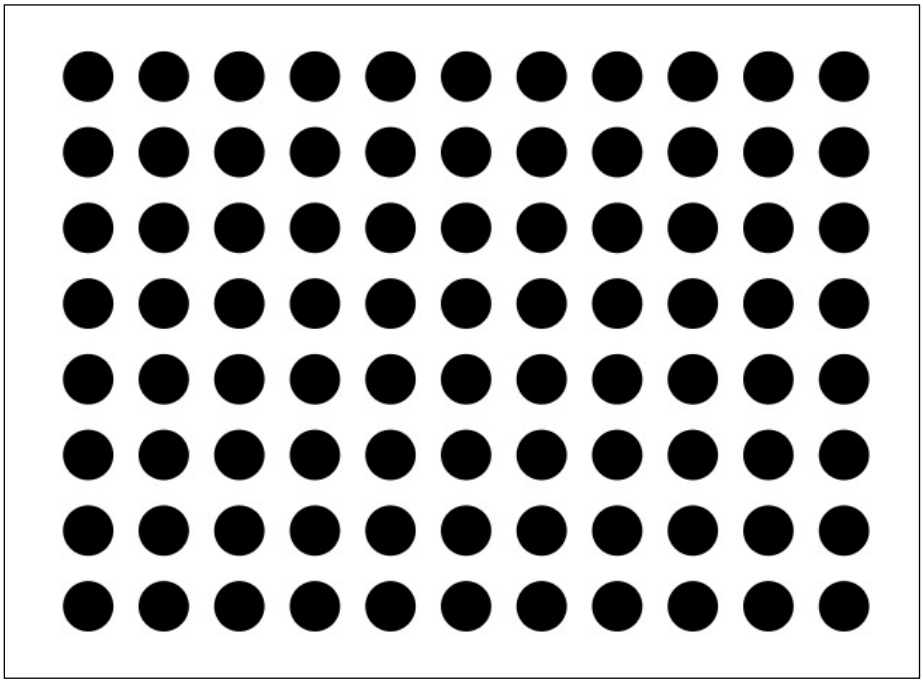


Figure 9. Example of circular style planar calibration board.

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Figure 10. Example of non-planar calibration fixture.

b. Calibration steps generally are as follows:

(1) Select calibration type appropriate for the analysis software and FOV.

(2) Capture time-synchronized calibration images in all cameras being used for analysis. For planar targets, different positions and orientations of the board filling up as much of the FOV of cameras and volume of space for measurement should be captured. Good practices are to cover at least 80 percent of the FOV, without the calibration points leaving the FOV, and for the calibration points to be laid in a randomized x, y, z pattern in the volume of interest, without being coplanar, as shown in Figure 10. When using non-planar calibration fixtures, a minimum of six calibration nodes is required for an algebraic solution. However, 8 to 12 calibration nodes are ideal as many software have algorithms built in to reduce the error in calibrations when more points are used. Depending on the software being used for calibration and analysis, upwards of 20 points may be recommended.

(3) Record focal length and camera resolution for analysis software input. Some analysis software may also require pixel pitch. Lens settings, camera position, and camera orientation must not be changed once calibration imagery has been acquired.

(4) Run through calibration software to calculate camera intrinsic and extrinsic parameters.

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(5) Examples of planar calibrations taken using only a calibration board are shown in Figure 11. An ideal calibration image and, depending on the software, is usually part of a greater image set as shown in Figures 12 and 13. This set of calibration images contain all the different calibration board orientations used to characterize the entire FOV. A non-ideal calibration image as the board does not sufficiently cover a significant portion of the FOV, nor would it be possible to move the board around to do so at its current standoff distance without additional equipment (Figure 12). Nonetheless, depending on the test, these types of calibration images may be the only ones viable. If necessary, walk, or otherwise move, the calibration board across as much of the FOV as possible. Understanding the limitations of the acquired calibration solution is paramount. A nonideal calibration image as the calibration board is not completely in frame as shown in Figure 13; these images are usually taken out of a calibration image set.



Figure 11. Ideal coverage of calibration board.

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Figure 12. Poor coverage of FOV.



Figure 13. Calibration board outside of frame.

(6) Ideal non-planar calibration views for a synchronized camera pair are shown in Figures 14 and 15. The calibration nodes are positioned in a random array while covering a majority of the FOV, ensuring quality calibration for an absolute orientation setup. In addition, a calibration board was also added to serve as a reference dimension should the calibration images be used in a relative orientation setup instead of absolute orientation.

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Figure 14. Ideal calibration view for a camera pair.

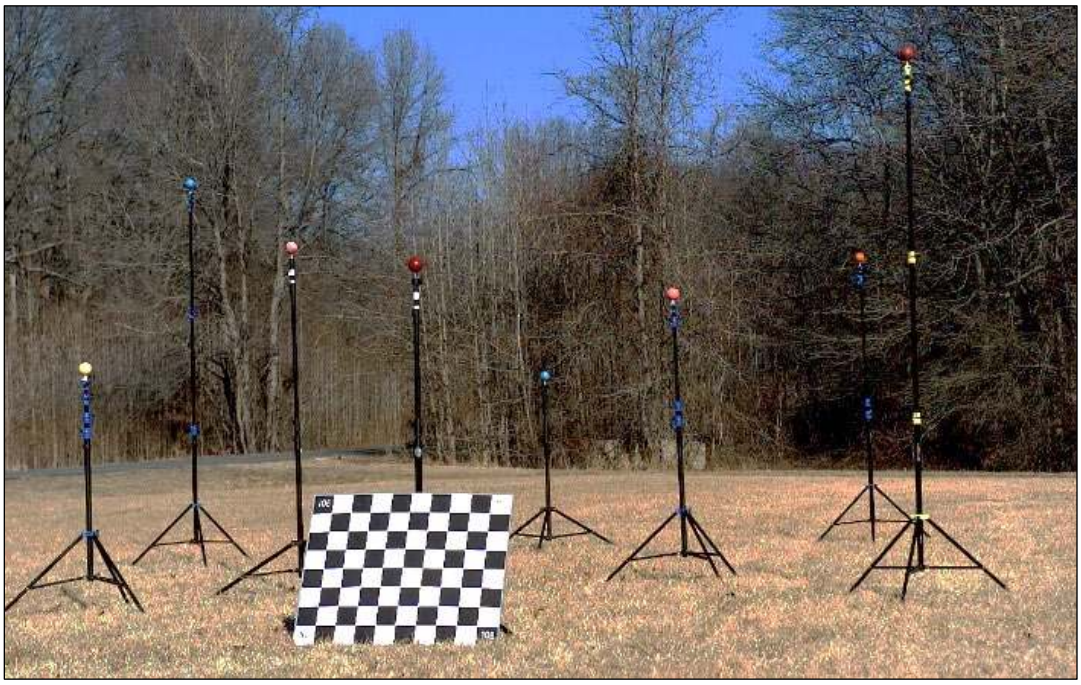


Figure 15. Ideal calibration view for a camera pair.

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(7) Examples of poor calibration views for a synchronized camera pair are shown in Figures 16 and 17. The camera view in Figure 17 is significantly more zoomed in on the calibration nodes than in Figure 16. This causes a discrepancy in which calibration nodes are visible in each camera view for correlation, thus altering the final calibration.

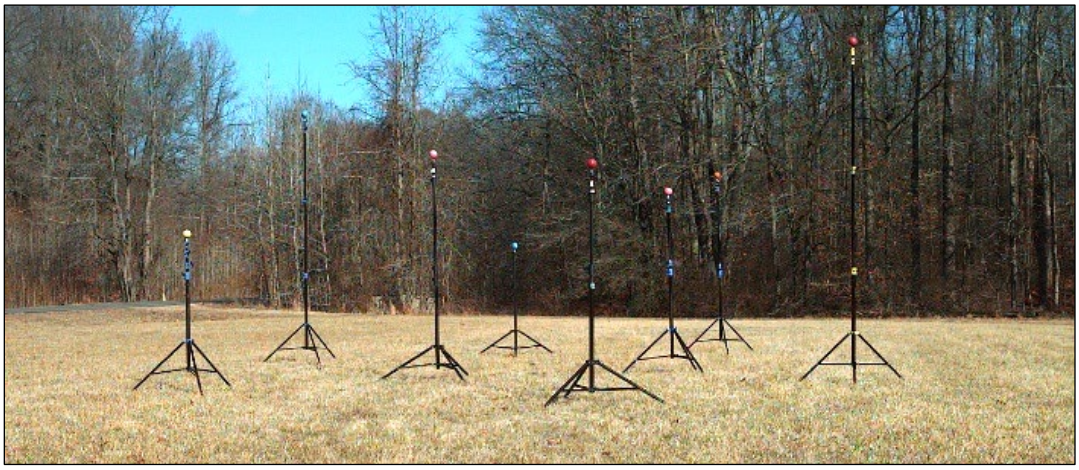


Figure 16. Non-ideal calibration views for a camera pair.



Figure 17. Non-ideal calibration views for a camera pair.

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(8) Additional examples of poor calibration setups can be seen in Figures 18 and 19. In Figure 4.18, the camera is focused on the front most calibration node in red. This results in the rest of the calibration fixtures, in the volume of interest, behind it being outside the depth of field. Thus, appearing soft/blurred. A camera should be focused on the center of the volume of interest, and camera aperture should be adjusted to make sure all fixtures in the volume of interest are in focus. All the calibration nodes are positioned in a straight line as shown in Figure 19. Poor sampling of the area of interest's volume and poor coverage of the camera's FOV would be the result. Both calibrations will lead to inaccurate 3D solutions during analysis.



Figure 18. Example of poor depth of field.



Figure 19. Example of poor placement of calibration nodes.

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(9) It is highly recommended to obtain a ground truth measurement (a known distance in the FOV that is independent of the calibration) before acquiring test data. This ground truth measurement can be used to verify the validity of your calibration. If calibration results are not optimal, adjustments and recalibration might be needed before acquiring test data. Adjustments may include changing camera positions, changing calibration fixture positions, adjusting focal length, FOV, etc.

4.4 Acquisition.

Data acquisition will be done using the appropriate high-speed camera control software. Proper settings should be checked before acquisition. Examples include camera synchronization configuration, timestamps, trigger input, exposure, and frame rates. Frame rates should match between all high-speed cameras used during testing for analysis. Capture imagery data and download the raw files and associated metadata. Compressed file formats such as MP4 are not ideal for data analysis in most cases as acquisition metadata is not captured and image quality is reduced.

5. DATA ANALYSIS.

a. This section will discuss steps for 3D analysis for different test configurations using Trackeye and ProAnalyst software packages. The two software are COTS software predominantly used by ATEC ranges for the majority of 3D image analysis. However, there are other software packages and custom programs, developed by test centers, that provide similar capabilities. The general processes discussed below can be applicable to other software packages as well.

b. Most software tools should have calibration, preprocessing, manual or automatic feature/object tracking, triangulation, and data visualization process. A generalized 3D workflow of Trackeye and ProAnalyst will be discussed below.

5.1 Project Setup.

a. The first step in the image analysis process is to set up a new project. This space will be where assets such as calibration and test video, survey data, target models, and camera definitions will be organized and managed.

b. To start a new project, you need to set up a project archive in Trackeye or create a New Project in ProAnalyst. In Trackeye, a project archive is a database that stores all the data associated with the 3D project, organized in a folder structure like the Windows file system. Camera definitions, calibration sessions, and tracking session are stored within the project archive (Figure 20). A session in Trackeye is a self-contained environment where data is loaded, analyzed, and processed.

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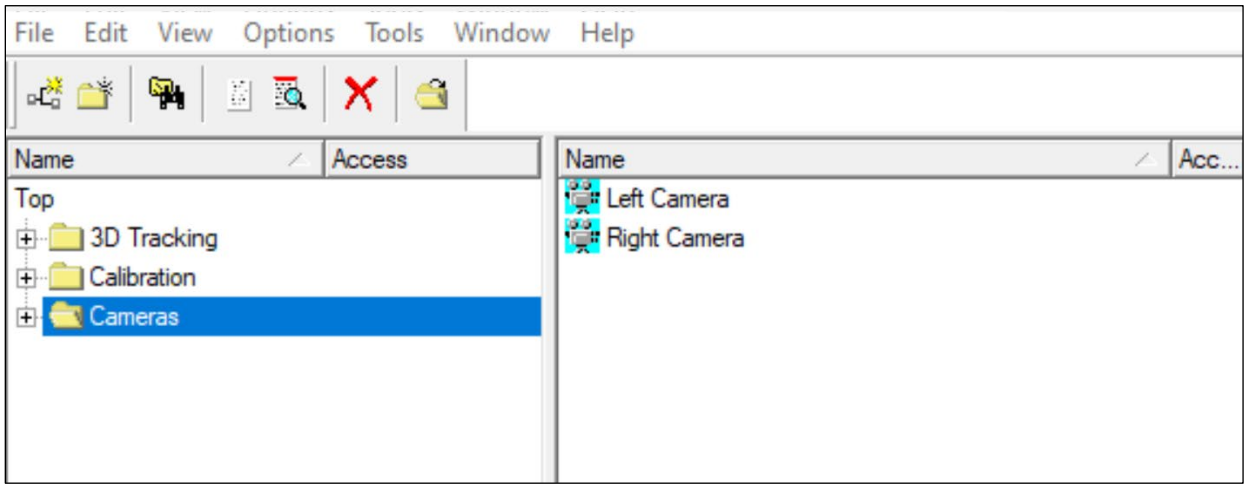


Figure 20. Trackeye Project archive.

c. In ProAnalyst, similarly, the project contains all the videos, associated calibrations, tracking, and analysis data. For 3D analysis, a 3D manager window is used for each stereo pair. 3D manager has tabs for: Manager, Calibration, Measurement, and World Reference. To use data from multiple camera pairs, each pair is analyzed independently using the 3D Manager, then combined using the 3D Merge (Figure 21).

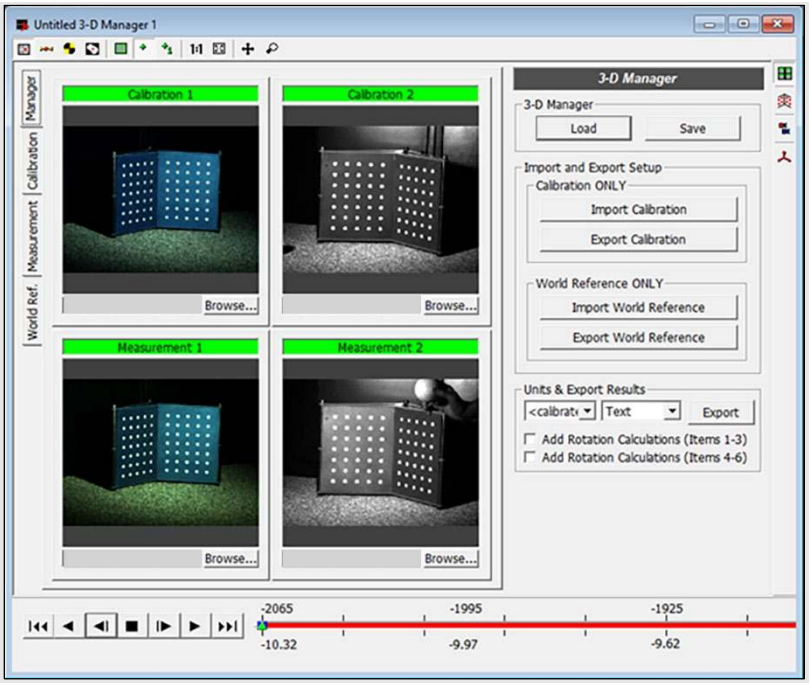


Figure 21. ProAnalyst 3D Manager.

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5.2 Calibration.

The two calibration methods for fixed cameras using ROC and AOC are discussed in section 4.1.2. The calibration process for the two methods on the software side will be discussed below. With both AOC and ROC methods, it is common to record the calibration footage before the test event. It is good practice to capture distinct locations such as aim point markers or other features in the FOV (e.g., sharp corners, fiducial markers in the foreground or background) that will be visible in the test footage since calibration fixtures are removed during the test shot. These known points captured in the calibration footage can be used to verify and correct accidental camera movements between calibration and testing by comparing their pixel location.

5.2.1 Relative Orientation Calibration (ROC).

a. Relative orientation calibration designates one camera as the coordinate system origin and calculates relative position and orientation of the second camera to that of the first. Calibration needs to be done in pairs if more than two cameras are used in testing.

b. ROC in ProAnalyst is performed according to the following steps: load a fixture file, locate image points within the calibration images, align the points to the fixture file, and calibrate cameras. A fixture file is a file that contains a collection of 3D cartesian coordinates as well as the unit system being used. ProAnalyst has two types of image point location, automatic and manual. Xcitex, the developer of ProAnalyst, offers calibration fixtures that the software will recognize and automatically locate the image points once boundaries have been established as shown in Figure 22. Associated with these Xcitex calibration fixtures are standardized fixture files depending on the size of the fixture purchased. When not using a calibration fixture from Xcitex, the user creates a custom fixture file that contains known cartesian coordinates that are visible in both camera views. These known coordinates are then manually selected in each of the camera views and aligned with the custom fixture file. ProAnalyst gives the user the option to view the calibration points in 3D space, this is illustrated in Figure 23. After the image points have been aligned with the fixture file, the cameras are calibrated, and the user may view the extrinsic and intrinsic parameters of each camera in the pair, as shown in Figure 24. The scene is now considered calibrated. As discussed in section 4.3, it is best practice to verify calibration using ground truth.

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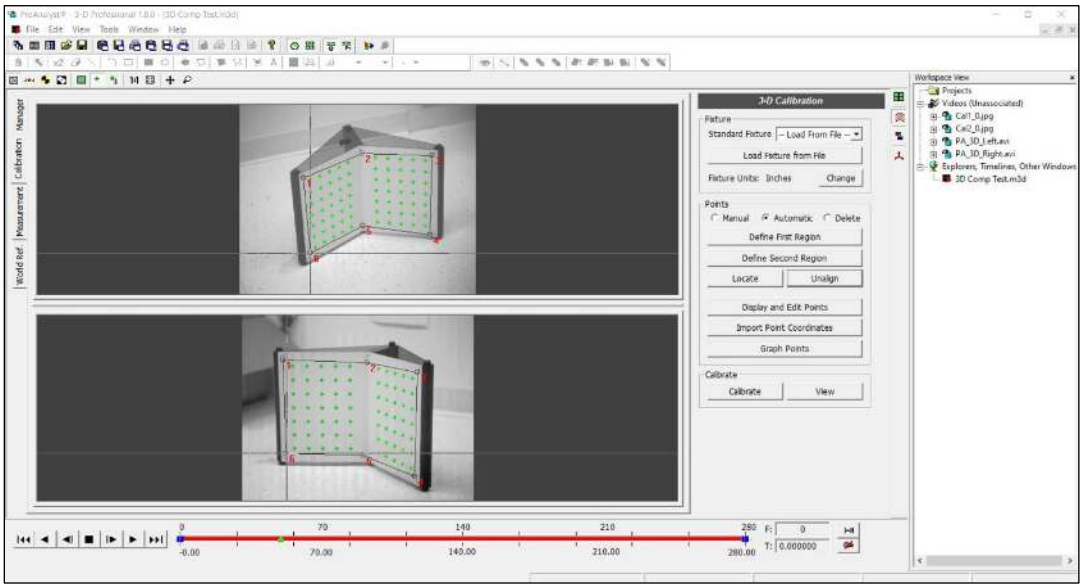


Figure 22. Calibration screen in ProAnalyst.

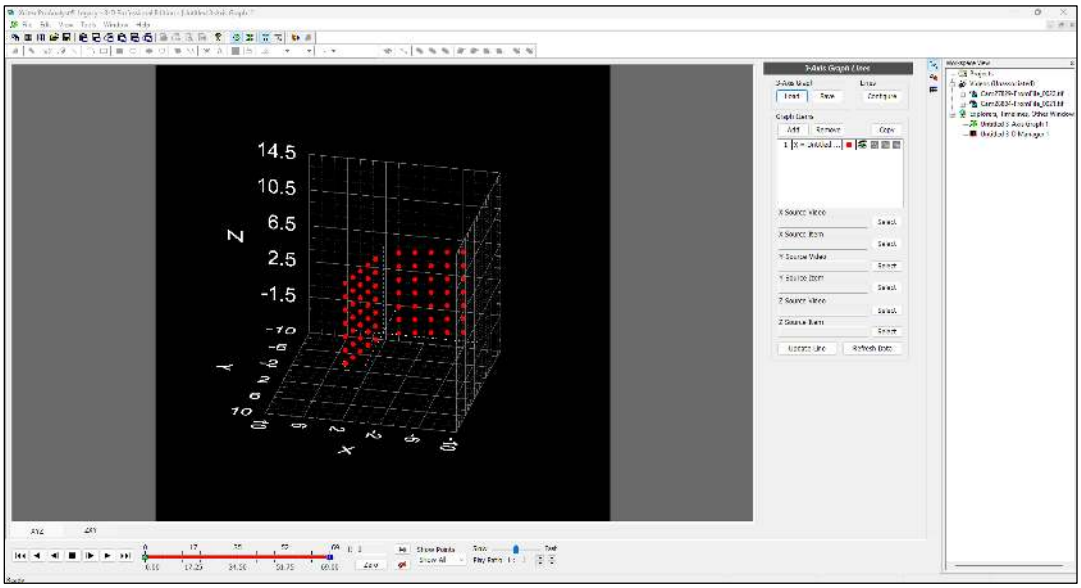


Figure 23. 3D calibration window and result.

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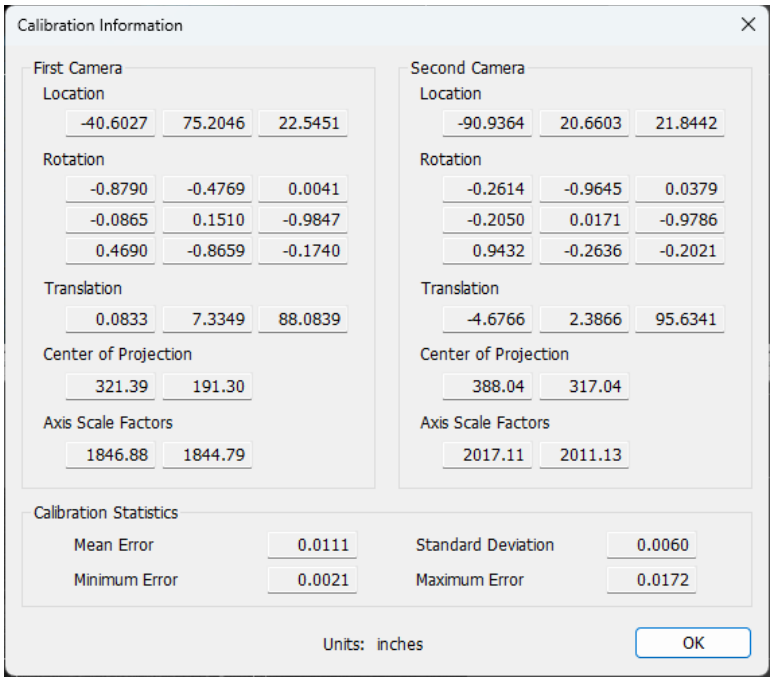


Figure 24. 3D calibration result.

c. ROC in TrackEye is performed according to the following steps:

(1) Create cameras in the archive and input the resolution, pixel size, and focal length of each camera lens as shown in Figure 25.

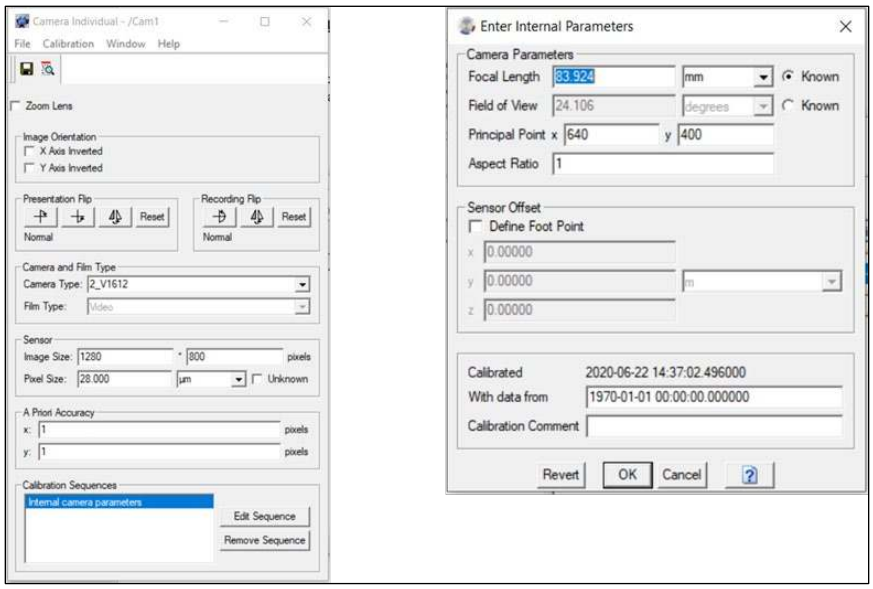


Figure 25. Camera and sensor definition window.

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(2) Create a new session. Place two Load Image icons followed by two 2D Tracking icons, then connect them with a Merge 2 icon followed by a ROC, shown in Figure 26.

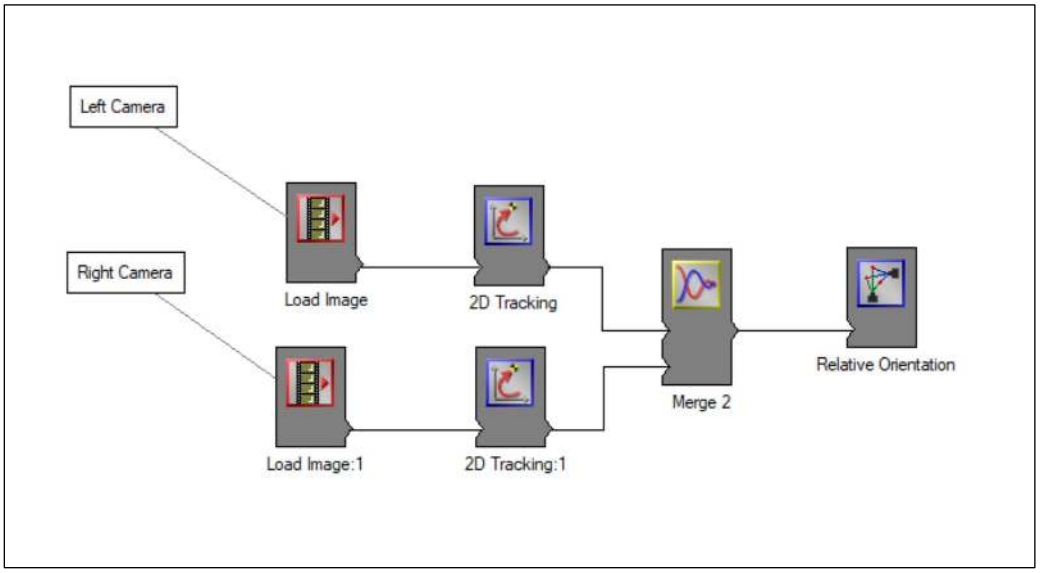


Figure 26. ROC session.

(3) Add calibration video files in the Load Image nodes. Open both 2D tracking windows for each camera view (Figure 27) and add a point for every location in the image to be used for calibration. Locations should be visible in both camera views. Eight- to 12-matching points in each view must have the same point names. At least two points in each camera view need to have a known distance between each point.

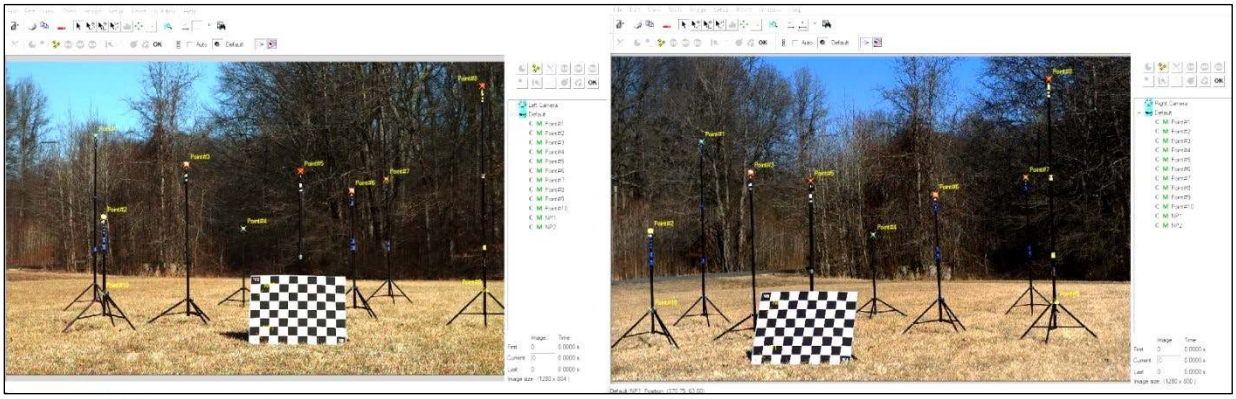


Figure 27. 2D Tracking window with calibration points.

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(4) In the ROC node, set the two points with the known distance, cameras to calibrate, and which camera to use as the primary or coordinate origin. Calibrate and check the root mean square (rms) parallax (Figure 28). The rms parallax is the rms value of the individual point parallaxes. Check to make sure rms parallax and residuals are less than or equal to the desired measurement accuracy for testing. Adjustments can be made to correct individual point placement in the 2D image. Camera parameter definition or lens distortion can also be a source of error. Verify the baseline (distance between the two cameras) and roll, pitch, yaw of second camera relative to the first camera makes physical sense.

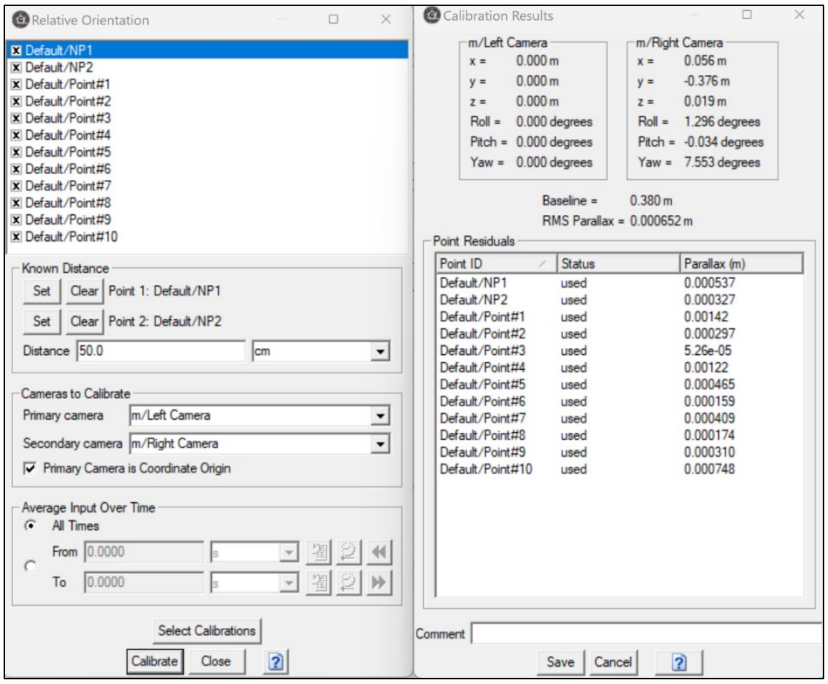


Figure 28. Relative orientation module.

5.2.2 Absolute Orientation Calibration (AOC).

a. AOC determines cameras position and orientation within an absolute world coordinate system. Typically, this coordinate system comes from geodetic survey points in cartesian x, y, z coordinate system of the camera location and calibration fixtures. These coordinate systems are imported as custom fixture files in ProAnalyst. In TrackEye, they are called a Target Model.

b. AOC in ProAnalyst is performed following the steps described in section 5.2.1 but with a fixture file that has the surveyed calibration fixture location points in the world coordinate system (Figure 29).

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```
% units: meters
% number of points
12
% Point Coordinates

-4.05, -0.85, 1.12
-3.48, 1.80, 2.25
-2.64, 0.00, 2.00
-0.5, 2.10, 1.30
0.00, 0.00, 1.76
1.00, 0.75, 1.50
1.87, 1.98, 1.80
2.31, 0.25, 2.21
```

Figure 29. Custom fixture file for AOC in ProAnalyst.

c. AOC in TrackEye is performed according to the following generalized steps:

(1) Create cameras in archive and input the resolution, pixel size, and estimated focal length of each camera lens.

(2) Create a Cartesian Target Model in the archive and manually add or import from a .csv file, the surveyed calibration points and camera locations (Figure 30).

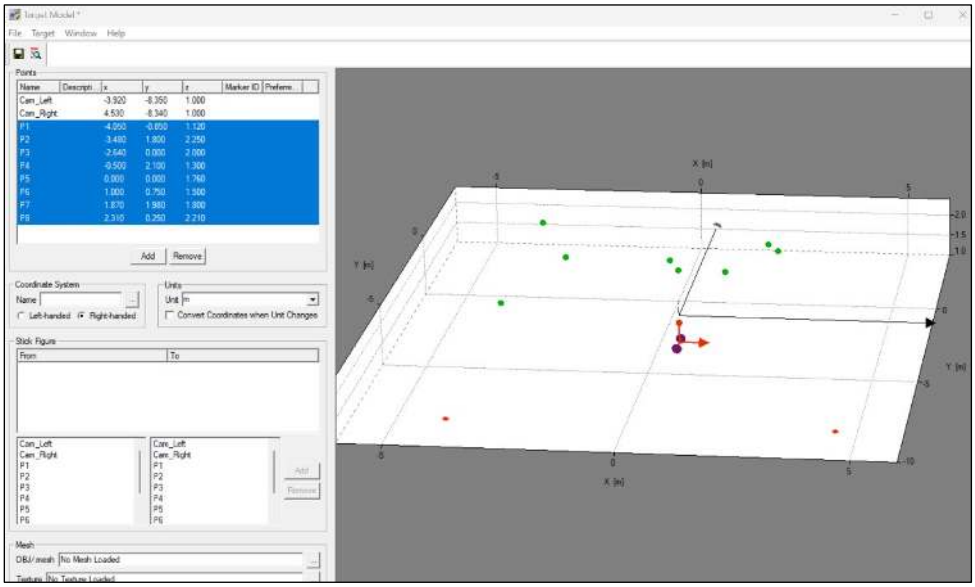


Figure 30. Target model import window. Cameras (red) and calibration fixtures (green).

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(3) Add calibration video files in the Load Image nodes. Open both 2D tracking windows for each camera view and select Add Target from Archive button. This opens the target model created in step 2. Select the calibration points, camera locations, and load them into the 2D tracking window.

(4) Place all visible calibration points in the corresponding location in the image as shown in Figure 27. Use Point Zoom button to zoom in and ensure that all the points are placed as accurately as possible. (The camera location point should be turned off as it is not visible in the image).

(5) In the AOC node, scroll through the Reference Points list and deselect the point for the camera, then ensure the correct camera is selected from the drop down next to Camera to Calibrate. Under Camera Position, use the drop-down menu to select the camera from the target model. Select Calibrate Focal Length and Calibrate Position optionally (calibrating these is not necessary if they are known in advance, but in some cases, it can be useful to verify recorded focal length and camera location). After everything is selected appropriately, click Calibrate (Figure 31).

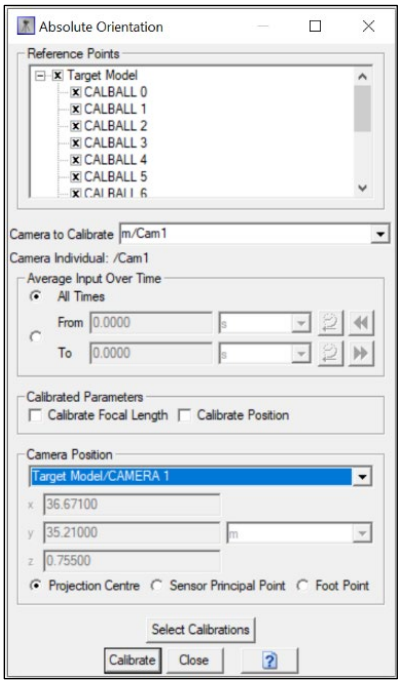


Figure 31. AOC calibration session.

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(6) The AOC results window will be displayed as shown in Figure 32. Check the value of rms Residual (similar to rms Parallax as discussed in section 5.2.1) to ensure the calibration is satisfactory for the test. If the rms residual is high for the desired test accuracy, check the individual point residuals for errors in point placement or survey data and either make corrections to those points in the 2D tracking window, target model, or delete them if there are enough calibration points. A minimum of 8 to 12 points is recommended for a calibration solution; however, more points are preferred. Save and close the calibration session.

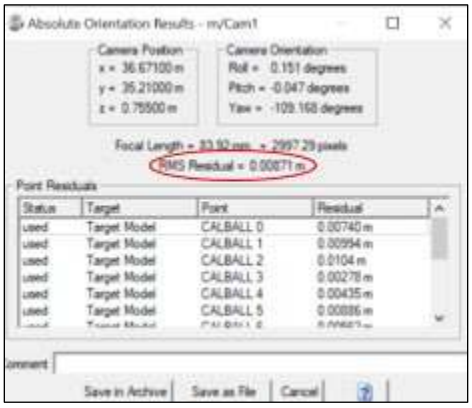


Figure 32. AOC calibration session.

(7) After the cameras are calibrated, if required, an orientation snapshot can be taken, and the pixel coordinates of the calibration objects can be manually input into the tracking window via the Tracker Setup window.

5.3 Tracking.

a. In ProAnalyst, each video file is independently tracked to produce a set of x, y image coordinates that are subsequently combined and translated into 3D world coordinates. Each 2D tracking window has the user select a region of interest from which the programs algorithms then attempt to find in subsequent frames, thus allowing for automatic tracking. Various settings that can be altered within an automatic track to improve accuracy and track stability (Figure 33). However, if the region of interest changes significantly, is not well defined, or has a changing background, automatic tracking can be difficult. Manual tracking is always an option in ProAnalyst if automatic tracking proves to be challenging due to issues such as the feature being tracked changing shape or not having enough pixel value separation from the background (Figure 34). After all items of interest have been tracked in each camera view, ProAnalyst has two options for merging your points, both with the same results. First, add where you select one point and then select its matching pair. Second, add matching where you select one point and ProAnalyst automatically selects its matching pair by assuming the matching pair has the same name.

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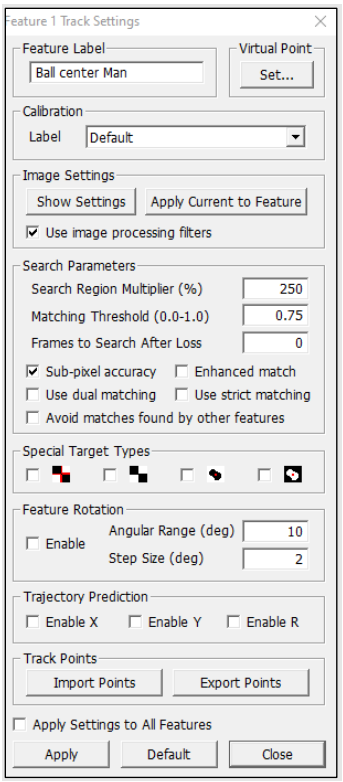


Figure 33. Feature tracking setup window.

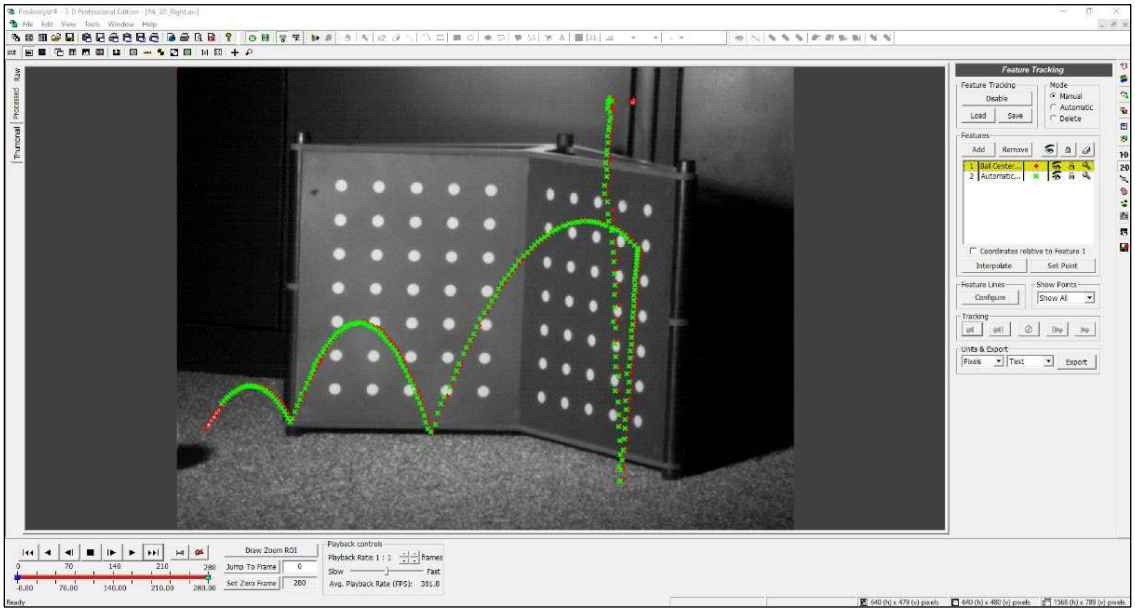


Figure 34. 2D tracking using both automatic tracking (green) and manual tracking (red).

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b. In Trackeye, each 2D tracking module is used to track objects in each video independently and paired with another to triangulate 3D position. For each point being tracked, the Point Setup window (Figure 35) is used to configure tracking parameters. This window allows the user to define how tracking is carried out for a selected point. The user can select features such as automatic versus manual tracking, tracking mode, tracking template size, and tracking tolerance for each point being tracked.

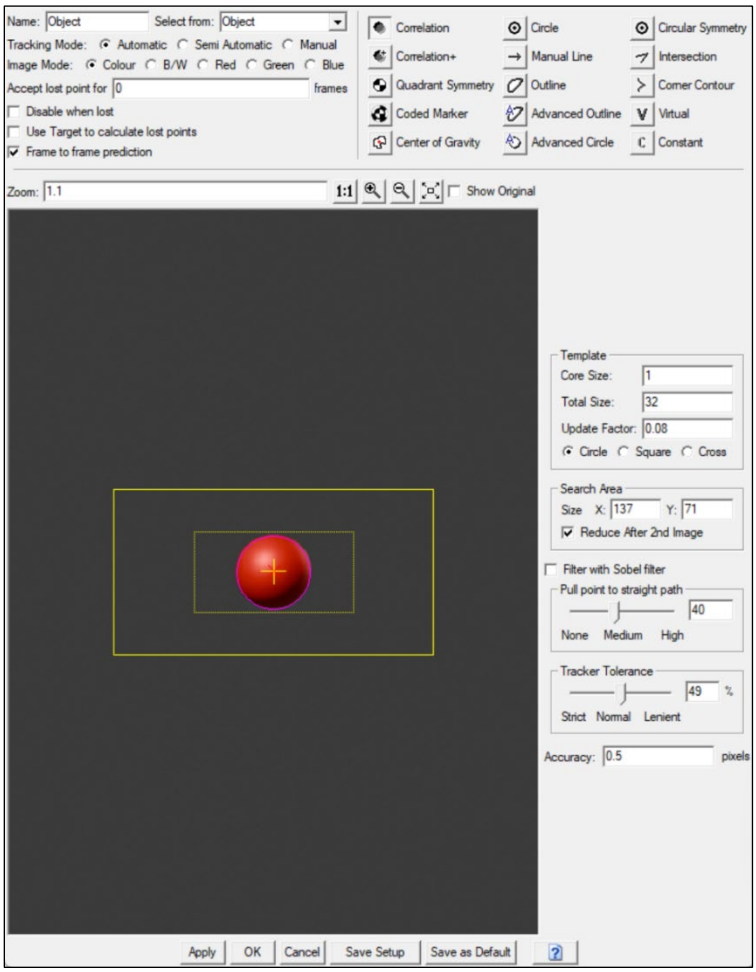


Figure 35. Point setup window.

c. After tracking is completed, the next step will be to perform all the necessary post processing such as performing kinematic calculations of the object being tracked. The different measurements and calculations can then be visualized in a tabular and plot formats.

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6. PRESENTATION OF DATA.

TrackEye and ProAnalyst software packages have tabular, 2D, and 3D display of TSPI analysis data. Examples of ProAnalyst and TrackEye tabular data are shown in Figures 36 and 37. Data presentation can be configured in a variety of ways to display specific points, calculations, axis, and coordinate systems. Examples of 3D plots of tracked data are shown in Figures 38 and 39. Dataset can be exported as .csv and .mat file formats to be further post-processed and visualized in other software such as Excel, Matlab, and Python.

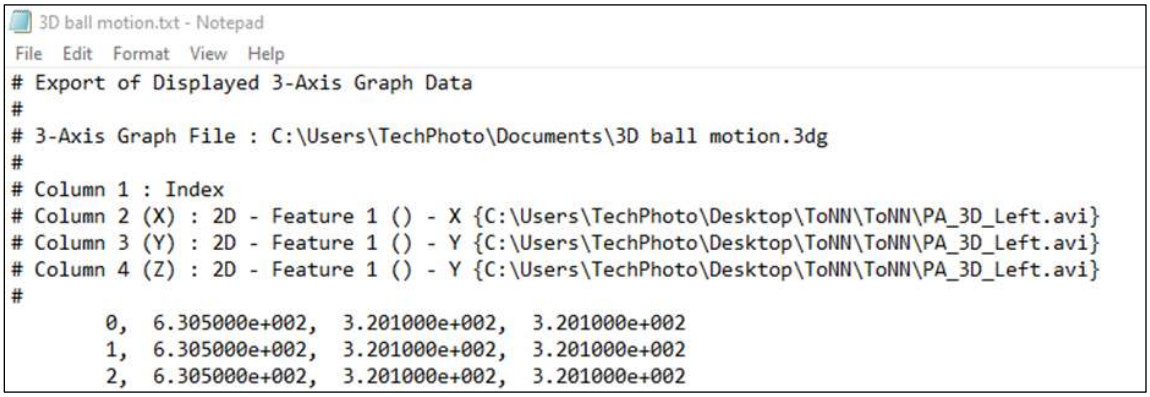


Figure 36. Tabular display of tracked data in ProAnalyst.

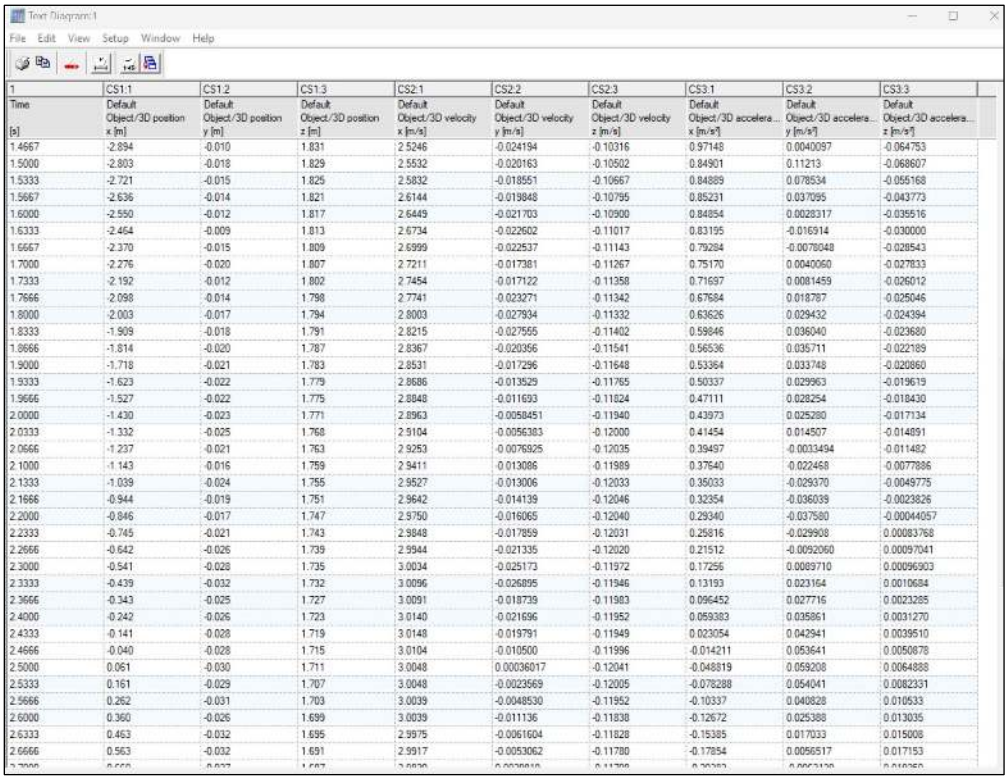


Figure 37. Tabular display of tracked data in TrackEye.

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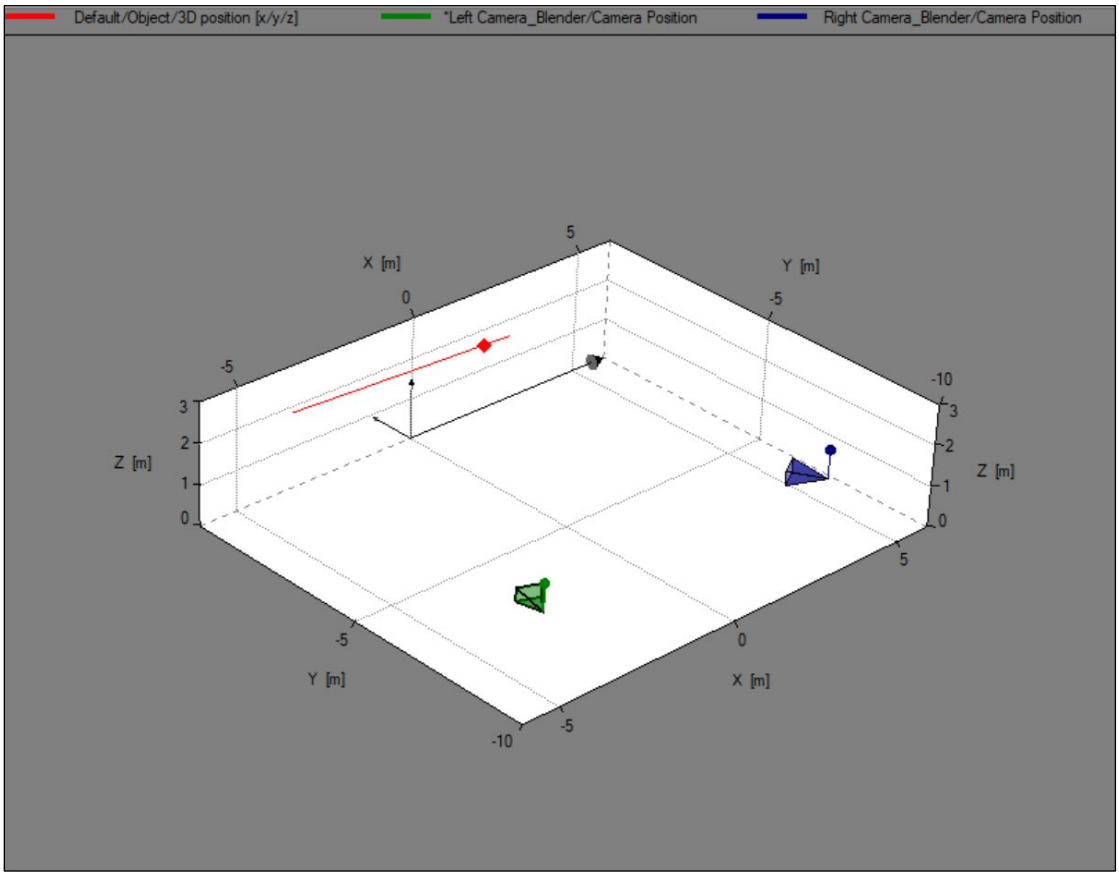


Figure 38. 3D plot of tracked data in TrackEye.

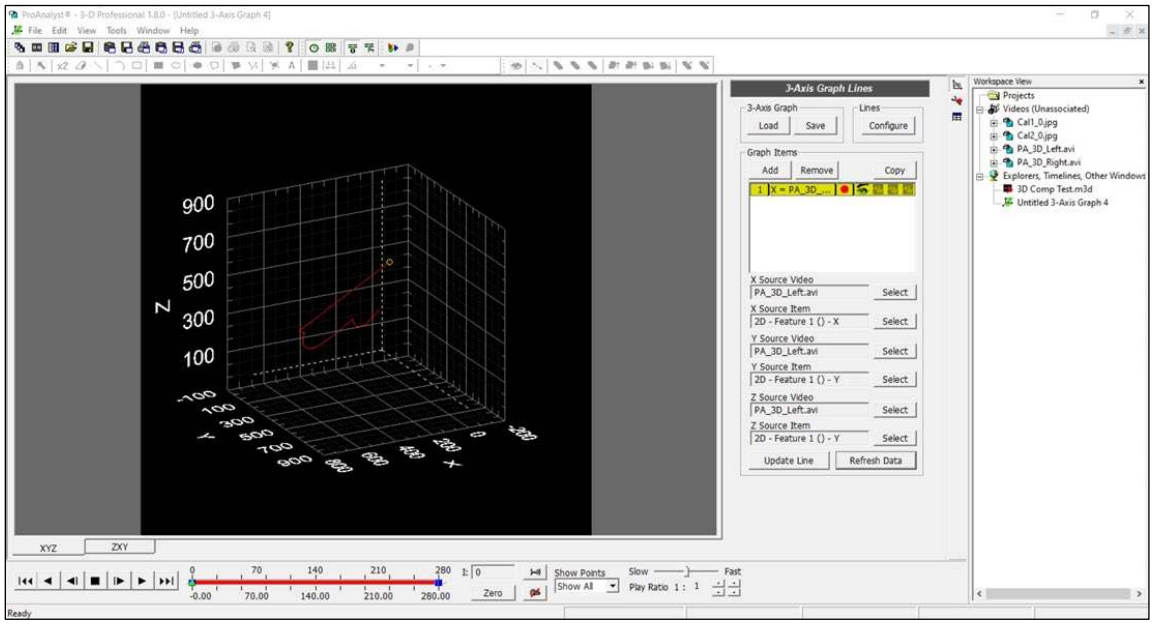


Figure 39. 3D plot of tracked data in ProAnalyst.

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APPENDIX A. ABBREVIATIONS.

2D	two-dimensional
3D	three dimensional
AOC	absolute orientation calibration
APS	Active Protection System
ATEC	U.S. Army Test and Evaluation Command
COTS	commercial off-the-shelf
FOV	field-of-view
fps	frames per second
GPS	Global Positioning System
IRIG-B	Inter-Range Instrumentation Group Format B
LED	light-emitting diode
LOS	line-of-sight
NFOV	narrow field-of-view
PIP	Projected Intercept Point
rms	root mean square
ROC	relative orientation calibration
SOP	Standard Operating Procedure
T&E	test and evaluation
TOP	Test Operations Procedure
TSPI	Time-Space-Position
WFOV	Wide field-of-view

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APPENDIX B. REFERENCES.

1. Xcitex ProAnalyst User Guide, Copyright 2003-2021, Xcitex, Inc.

2. TrackEye User's Guide, Copyright 2023 Image System AB.

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SUBJECT: Test Operations Procedure 01-2-100 3-DIMENSIONAL IMAGE ANALYSIS USING HIGH-SPEED CAMERAS Publication

1. Test Operations Procedure (TOP) 01-2-100 3-DIMENSIONAL IMAGE ANALYSIS USING HIGH-SPEED CAMERAS has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. The objective of this TOP is to cover three-dimensional (3D) image analysis techniques using fixed high-speed cameras to support test and evaluation (T&E) programs.

3. This document is approved for publication and will be posted to the ASSIST website. The ASSIST website can be accessed at https://assist.dla.mil/.

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